

Assignment 4 DCCN

Submit on or before 30/09/2026

- 1. Given the IPv4 address 180.20.10.20/20, determine the following:**
 - (i)** Class of the IP address
 - (ii)** Default subnet mask
 - (iii)** Custom subnet mask
 - (iv)** Number of subnets created
 - (v)** Number of usable host addresses per subnet
 - (vi)** Valid subnet (network) addresses
 - (vii)** Broadcast address of each subnet
 - (viii)** Network address and broadcast address of the subnet to which **180.20.10.20** belongs
- 2. Define congestion control. Explain the principles of congestion preventions and control.**
- 3. Draw the schematic diagram of IPV4 header. Compare between IPv4 and IPv6.**
- 4. Explain the TCP segment format with a neat diagram. Compare TCP and UDP based on their major features.**